

SPRiF: Spectral Projective Reset Integrate-and-Fire Neuron

Anonymous submission

Abstract

In standard spiking neurons, the membrane potential accumulates input, triggers spikes, and receives the post-spike reset. This structural coupling makes the state that carries temporal context also absorb every reset. We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), which separates these functions through *functional state decomposition*. A constrained spectral slow state combines a real decay mode with a two-dimensional damped rotation and is never reset by spikes; a fast discharge state produces the membrane readout and receives a learnable projective reset along $[1, \lambda]^\top$. Across S-MNIST, PS-MNIST, QTDB, Google Speech Commands (GSC), and Spiking Heidelberg Digits (SHD), SPRiF obtains the highest reported accuracy in our comparison on three benchmarks (99.28%, 95.86%, and 88.43%) and remains competitive on GSC (94.55%) and SHD (91.52%). Mechanism ablations show that removing rotation, merging the slow and fast states, and fixing the reset to a scalar direction reduce accuracy by up to 3.53, 5.04, and 1.03 points, respectively. State trajectories directly verify that spike-triggered discontinuities are confined to the fast state, while learned impulse responses exhibit heterogeneous decay and oscillatory kernels. These results support functional state decomposition as a practical neuron-level design principle for temporal SNNs.

1 Introduction

Spiking neural networks (SNNs) process temporal information through discrete spike events, offering an event-driven computational paradigm with applications in speech recognition, physiological signal analysis, and event-based recognition. Unlike continuous-activation networks, the representational capacity of an SNN depends not only on its inter-layer connectivity but critically on the internal dynamics of individual neurons: how each neuron accumulates input history, decides when to fire, and reorganizes its state after each spike event. For temporal tasks where the relevant information is distributed across multiple timescales—from milliseconds in speech phonemes to hundreds of timesteps in long permuted sequences—the design of neuron-level dynamics directly constrains what temporal patterns the network can retain and exploit.

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The leaky integrate-and-fire (LIF) neuron remains the dominant computational unit in SNNs due to its simplicity and compatibility with surrogate-gradient training. A rich lineage of extensions has substantially enriched the LIF dynamics: adaptive threshold models introduce slow auxiliary variables that modulate firing behavior (Baronig et al. 2025; Zhang et al. 2025b; Idrees et al. 2024); resonant and oscillatory neurons add damped subthreshold oscillations for frequency-selective processing (Higuchi et al. 2024; Huang et al. 2024b; Zhang et al. 2025a); multi-timescale designs equip neurons or synapses with heterogeneous decay rates (Wei et al. 2026; Deng et al. 2025; Fullea-Garcia, Soldado-Magraner, and Margarit-Taulé 2026; Tiwari and Chauhan 2025); and recent reset mechanism innovations redesign the post-spike state update to reduce information loss (Deng et al. 2022; Huang et al. 2025; Li et al. 2025; Fang et al. 2023). Despite this diversity, all existing spiking neurons share a structural assumption: **the state variable that triggers a spike is the same state variable that gets reset**. In standard LIF, the membrane potential simultaneously accumulates past input, determines spike timing, and absorbs the reset—meaning every spike event partially erases the neuron’s temporal memory. Even neurons that add oscillatory dynamics (e.g., resonate-and-fire variants (Higuchi et al. 2024; Zhang et al. 2025a)) or adaptation variables (e.g., SE-adLIF (Baronig et al. 2025)) still reset the membrane potential that serves as the primary readout. Recent reset innovations—whether they soften the reset (Deng et al. 2022), make it adaptive (Huang et al. 2025), model it via threshold dynamics (Li et al. 2025), or eliminate it entirely for parallelization (Fang et al. 2023)—all operate on, or reason about, a single state that simultaneously stores memory and generates spikes.

This structural coupling motivates the central question of this work: **must the spike-triggering state and the memory-bearing state be the same variable?** From a temporal modeling perspective, a spike is a discrete event signaling that the current state has reached a threshold; it necessitates a correction of the discharge-related state, but it does not logically require that all accumulated temporal memory be simultaneously weakened or erased. If the memory-carrying state could be structurally insulated from spike-triggered reset, the neuron would retain a richer record of input history across spike events—potentially enabling more faithful tem-

poral processing without increasing network size or abandoning binary spike communication.

We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), a spiking neuron that addresses this question through *functional state decomposition*. SPRiF separates the neuron into two dynamically distinct components: a **slow spectral memory state** and a **fast discharge state**. The slow state integrates input through a constrained spectral structure—comprising a real exponential decay mode and a two-dimensional damped rotation mode—providing structured temporal filtering that extends beyond the single-exponential LIF memory trace. By design, the slow state is **never reset** by spike events, preserving continuous temporal memory across the spike train. The fast state receives projections from the slow state, produces the membrane readout, generates binary spikes via threshold crossing, and absorbs a novel *projective reset* along a learnable direction $[1, \lambda]^\top$ in its two-dimensional state space. SPRiF is fully compatible with standard SNN training: it uses backpropagation through time (BPTT) with surrogate gradients, requiring no specialized optimization. This design realizes what we term the *spike-never-resets-memory* principle: spike events trigger corrections only on the fast discharge manifold, while the slow spectral memory evolves continuously and undisturbed.

We emphasize that SPRiF’s functional decomposition is qualitatively different from existing multi-timescale neuron designs. Prior multi-timescale models assign different decay rates or time constants to parallel processing channels—a strategy we term *rate decomposition*—but all channels serve the same dynamical role (memory accumulation and readout) and are subject to the same reset. SPRiF instead assigns different *dynamical functions* to its two states, enforcing the separation structurally through the spectral constraint on the slow state and the projective reset on the fast state, rather than through heterogeneous time constants alone.

We evaluate SPRiF across five temporal processing benchmarks spanning sequential image classification (S-MNIST, PS-MNIST), physiological signal analysis (QTDB ECG), speech recognition (Google Speech Commands), and event-based digit recognition (Spiking Heidelberg Digits). SPRiF achieves the highest accuracy in our comparison on S-MNIST (99.28%), PS-MNIST (95.86%), and QTDB (88.43%), and ranks second on GSC (94.55% vs. 94.84%) and SHD (91.52% vs. 91.70%). Its parameter count is lower than the strongest compared result on four datasets and comparable on QTDB. Mechanism ablations across PS-MNIST, GSC, and QTDB isolate the contribution of rotation, state separation, and projective reset. Complementary trajectory and impulse-response analyses expose how these components alter the neuron’s internal dynamics rather than treating SPRiF as a black-box replacement.

Contributions.

1. We propose *functional state decomposition* for spiking neurons—separating temporal memory storage, spike readout, and post-spike reset into dynamically distinct state components, a principle that is orthogonal to existing rate-decomposition and single-state-reset approaches.

2. We introduce a *constrained spectral slow state* with real exponential decay and damped rotation modes, providing interpretable, structured temporal filtering beyond the single-exponential LIF trace.
3. We propose *projective fast-state reset*, a learnable directional reset mechanism that operates exclusively on the discharge state and leaves the memory state untouched.
4. We provide empirical validation across five temporal benchmarks, three mechanism ablations, and dynamical analyses of state continuity and learned temporal kernels. Code will be released upon acceptance.

2 Related Work

2.1 Spiking Neuron Models Beyond Standard LIF

The leaky integrate-and-fire (LIF) neuron remains the dominant computational unit in spiking neural networks (SNNs), yet its single-variable membrane dynamics limit temporal expressiveness. Several extensions have been proposed: GLIF (Fang et al. 2022) and DGN (Bai, Wang, and Yu 2026) introduce gated integration and dynamic conductance mechanisms, respectively; ASRNN (Yao et al. 2021) and CLIF (Huang et al. 2024a) address adaptive thresholds and temporal gradient vanishing; and RadLIF (Kamata et al. 2022) develops surrogate-gradient baselines for speech tasks. Although these models enrich the LIF dynamics with gating, adaptation, or gradient-aware structures, they all share a common assumption: a single state variable serves simultaneously as temporal memory, spike readout, and reset target. SPRiF departs from this assumption by functionally decomposing the neuron into a slow spectral memory state that is never reset and a fast discharge state that handles readout, spike generation, and a learnable projective reset.

2.2 Adaptive and Multi-Timescale Neuron Dynamics

A parallel line of research recognizes that real neurons operate across multiple timescales and that single-timescale LIF dynamics are insufficient for tasks requiring both rapid responsiveness and sustained memory. The adaptive LIF family introduces slow auxiliary variables that modulate the fast membrane dynamics. AdLIF (Baronig et al. 2025) achieves state-of-the-art results on event-based benchmarks by adding a per-neuron adaptation variable inspired by biological spike-frequency adaptation; DA-LIF (Zhang et al. 2025b) extends this with dual spatial and temporal adaptation channels; and the broader study of biophysical adaptation mechanisms (Idrees et al. 2024) demonstrates how adaptation changes circuit computations to match prevailing input statistics.

Beyond adaptation, several recent works explicitly design multi-timescale processing into the neuron or its connections. Recurrent SNNs with bimodal neuronal time scales (Wei et al. 2026) introduce a dual-timescale architecture into spiking recurrent networks with specialized learning algorithms. HetSyn (Deng et al. 2025) relocates timescale diversity from the neuron membrane to individual synapses, assigning each synaptic connection a distinct decay rate. The Multi-Timescale Conductance (MTC) framework (Fulleda-Garcia, Soldado-Magraner, and Margarit-Taulé 2026) shapes

neural dynamics through fast, slow, and ultra-slow conductance channels. Local Timescale Gates (Tiwari and Chauhan 2025) equip each spiking cell with parallel fast and slow leaky integrators and a learnable blending parameter.

These multi-timescale approaches assign *different decay rates* to parallel state components—a strategy we term *rate decomposition*. SPRiF instead proposes *functional decomposition*: the slow and fast states serve distinct dynamical roles (continuous memory storage versus membrane read-out and discharge), and the separation is enforced structurally through the spectral constraint and projective reset, not merely through heterogeneous time constants.

2.3 Oscillatory, Resonate-and-Fire, and Dendritic Neuron Models

Biological neurons exhibit subthreshold oscillations that enable frequency-selective temporal processing. The resonate-and-fire (RF) model (Izhikevich 2001) formalizes this behavior through a complex-valued state variable with damped oscillatory dynamics. Recent work has substantially advanced the RF lineage. Balanced Resonate-and-Fire (BRF) neurons (Higuchi et al. 2024) introduce a balanced oscillatory mechanism that achieves higher task accuracy with fewer spikes and parameters. The Parallel Resonate-and-Fire (PRF) neuron (Huang et al. 2024b) decouples the reset operation from the complex-domain oscillatory dynamics to enable parallelized training. The Dendritic Resonate-and-Fire (D-RF) neuron (Zhang et al. 2025a) incorporates a bio-inspired multi-dendrite architecture for frequency-selective filtering.

Complementing the oscillatory direction, dendritic computation has been explored as a mechanism for enriching SNN expressiveness. TC-LIF (Yin et al. 2024) introduces a two-compartment model with spatially separated dendritic and somatic dynamics. DH-SNN (Zheng et al. 2024) incorporates temporal dendritic heterogeneity to learn multi-timescale dynamics. The Dendriify framework (Stevenson and Kappel 2023) provides a general toolkit for incorporating dendritic structures, and neural heterogeneity (Lange et al. 2021) has been shown to promote robust learning.

A critical limitation shared by oscillatory and dendritic models is that the oscillatory or dendritic dynamics reside in the same state—or are projected into the same membrane potential—that undergoes spike reset. SPRiF takes a different approach: the damped oscillation lives in a dedicated slow state that is *structurally insulated from reset*, preserving oscillatory phase and amplitude information across spike events. Furthermore, while TC-LIF and DH-SNN decompose the neuron spatially (dendrite versus soma), SPRiF decomposes it functionally (memory versus discharge).

A parallel line of work has explored importing structured state space model (SSM) techniques into spiking neurons (Fabre 2025; Shen et al. 2025). These methods enhance the training dynamics of single-state neurons; SPRiF instead redesigns the neuron’s internal state architecture from first principles, with oscillation arising intrinsically from spectral structure rather than from imported SSM parametrization.

2.4 Reset Mechanism Innovations

The post-spike reset mechanism is a defining feature of spiking neurons, yet standard hard reset—subtracting a fixed threshold value from the membrane potential—discards residual information. InfLoR-SNN (Deng et al. 2022) proposes a *soft reset* with Membrane Potential Redistribution to recover discarded information. RPLIF (Li et al. 2025) models the biological refractory period through spike-triggered threshold dynamics. AR-LIF (Huang et al. 2025) introduces an adaptive reset with a learnable mapping. At a more fundamental level, recent work (Zhang 2025) questions whether reset mechanisms are strictly necessary, and PSN (Fang et al. 2023) removes reset entirely to enable parallelized training.

All of these innovations operate on a **single state** that simultaneously stores memory and generates spikes. SPRiF introduces a qualitatively different solution: by structurally separating the neuron into a slow memory state and a fast discharge state, the reset operates exclusively on the fast state through a learnable *projective* direction $[1, \lambda]^\top$. The slow spectral memory is never touched by reset, achieving what we term *structural memory protection*—a property that no existing reset innovation provides.

2.5 Network-Level Architectures for Temporal Processing

Beyond neuron-level innovations, network-level designs have also advanced SNN capabilities. MPS-SNN (Ding et al. 2025b) reframes SNNs as ensembles of temporal subnetworks; stochastic gating (Ding et al. 2024) enhances adversarial robustness through probabilistic spike communication; and a comprehensive study (Ding et al. 2025a) demonstrates that temporal processing inherently enhances computational robustness. These contributions are complementary to SPRiF’s neuron-level functional decomposition.

3 Method

3.1 From Membrane-Centric Dynamics to Functional Decomposition

Let s_t denote the input to a layer at time t . A conventional LIF neuron maintains a scalar membrane state v_t , produces a binary spike $z_t = H(v_t - \theta)$, and applies a post-spike correction to the same state. Ignoring reset, its history is weighted by a single exponential kernel. SPRiF instead maintains a three-dimensional slow state $\mathbf{x}_t \in \mathbb{R}^3$ and a two-dimensional fast state $\mathbf{u}_t \in \mathbb{R}^2$. The slow state stores structured temporal context, whereas the fast state maps that context to a membrane potential and absorbs the reset. For a recurrent layer, the input current is

$$I_t = W_{\text{in}}s_t + W_{\text{rec}}\mathbf{z}_{t-1} + b, \quad (1)$$

where the recurrent term is omitted in feed-forward layers. Each neuron then follows the ordered computation $I_t \rightarrow \mathbf{x}_t \rightarrow \tilde{\mathbf{u}}_t \rightarrow v_t \rightarrow z_t \rightarrow \mathbf{u}_t$, where $\tilde{\mathbf{u}}_t$ is the pre-reset fast state. This ordering is important: the current slow state is computed before the spike, and reset never feeds back into it at the same step.

3.2 Constrained Spectral Slow State

The slow state contains one real decay mode and one damped rotational pair. Its update is

$$\mathbf{x}_t = A\mathbf{x}_{t-1} + BI_t, \quad (2)$$

with

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho \cos \omega & -\rho \sin \omega \\ 0 & \rho \sin \omega & \rho \cos \omega \end{bmatrix}, \quad B = \begin{bmatrix} 1 - \alpha \\ 1 - \rho \\ 0 \end{bmatrix}. \quad (3)$$

The first coordinate is therefore an exponential trace, $x_t^0 = \alpha x_{t-1}^0 + (1 - \alpha)I_t$, while (x_t^1, x_t^2) form a damped rotation driven through the first oscillatory coordinate. We parameterize

$$\alpha = \sigma(\alpha_{\text{raw}}), \quad \rho = \sigma(\rho_{\text{raw}}), \quad \omega = \pi\sigma(\omega_{\text{raw}}), \quad (4)$$

so that $\alpha, \rho \in (0, 1)$ and $\omega \in (0, \pi)$. This constraint gives each parameter a direct dynamical meaning and prevents the slow transition from becoming an unconstrained recurrent matrix.

The structure also makes the available temporal basis explicit. An input impulse contributes terms proportional to $(1 - \alpha)\alpha^k$ through the real mode and to $(1 - \rho)\rho^k \cos(k\omega)$ and $(1 - \rho)\rho^k \sin(k\omega)$ through the rotational pair after k steps. SPRiF can thus combine monotone and oscillatory traces within each neuron while retaining linear-time recurrence.

3.3 Fast Discharge State and Projective Reset

The slow state is translated into a two-dimensional pre-reset fast state by

$$\tilde{u}_t^0 = \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + (G\mathbf{x}_t)^0, \quad (5)$$

$$\tilde{u}_t^1 = \eta_1 u_{t-1}^1 + (G\mathbf{x}_t)^1, \quad (6)$$

where $\eta_0, \eta_1 \in (0, 1)$ are learned leak coefficients, κ couples the second fast coordinate to the first, and $G \in \mathbb{R}^{2 \times 3}$ projects slow dynamics into the discharge space. The membrane is the first coordinate, $v_t = \tilde{u}_t^0$, and the forward spike remains binary:

$$z_t = H(v_t - \theta). \quad (7)$$

We train through the discontinuity with the multi-Gaussian surrogate used in prior recurrent SNN training (Yao et al. 2021); the forward pass always uses the hard threshold.

After a spike, SPRiF corrects only the fast state:

$$\mathbf{u}_t = \tilde{\mathbf{u}}_t - z_t \theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}, \quad (8)$$

where λ_j is learned independently for neuron j . Setting $\lambda_j = 0$ recovers scalar subtraction on the membrane coordinate, whereas nonzero values let reset modify both discharge coordinates along a learned direction. Crucially, Eq. (8) contains no $\mathbf{x}_t$ term: a spike changes $\mathbf{u}_t$ but leaves the memory-bearing slow state intact. This structural property realizes the spike-never-resets-memory principle without removing reset or replacing binary communication.

3.4 Layer Construction, Training, and Cost

A SPRiF layer applies Eqs. (1)–(8) independently to all neurons and emits the spike sequence $[z_1, \dots, z_T]$. Layers can be stacked, and recurrence is optional rather than part of the neuron definition. Sequence-level tasks use a linear readout over temporally aggregated hidden spikes; QTDB uses a time-distributed classifier; SHD uses a spiking readout whose membrane evidence is accumulated after a short warm-up. All parameters, including $\alpha, \rho, \omega, G, \kappa, \eta$, and λ , are optimized jointly with network weights by backpropagation through time.

Relative to a scalar LIF neuron, SPRiF stores five state values and adds 13 neuron-wise dynamics parameters (three spectral parameters, two fast leaks, one coupling, six slow-to-fast projection coefficients, and one reset coefficient). The structured state updates remain $O(h)$ for a layer of width h ; when recurrent connectivity is enabled, the shared $O(h^2)$ matrix multiplication dominates both LIF and SPRiF variants. We report end-to-end parameter counts to expose this trade-off empirically.

4 Experiments

4.1 Benchmarks and Protocol

We evaluate whether one neuron design transfers across five temporal regimes. S-MNIST and PS-MNIST present 784 normalized pixels sequentially, with PS-MNIST applying one fixed random permutation. QTDB is a six-class ECG sequence-classification task. GSC uses 101-frame, 120-dimensional mel-spectrogram features for 12-class keyword spotting. SHD contains 700-channel event sequences and 20 spoken-digit classes. Accuracy is the evaluation metric throughout.

The S-MNIST and PS-MNIST networks contain two recurrent SPRiF layers of widths 64 and 256 and are trained for 150 epochs with Adam, batch size 512, learning rate 10^{-2} , and truncated BPTT chunks of 262 steps. QTDB uses one recurrent layer of width 36, Adam, batch size 128, learning rate 10^{-2} , and 250 epochs. GSC uses one recurrent layer of width 300, temporal mean pooling, AdamW, batch size 128, learning rate 5×10^{-3} , and at most 150 epochs. SHD uses a recurrent hidden layer of width 64 and a 20-neuron SPRiF readout, Adam, batch size 100, learning rate 10^{-2} , and at most 100 epochs. The corresponding seeds are 0, 0, 1111, 42, and 0. Spectral parameters are initialized from log-uniform time-constant ranges and then learned without clipping in their constrained parameterization.

We compare against published recurrent SNN results spanning gated neurons, adaptive neurons, dendritic models, resonate-and-fire models, and recent temporal architectures. Because the source studies use task-specific protocols, Table 1 reports the available single-model accuracies rather than claiming statistical significance across papers. The current SPRiF entries are single-run values from the task-specific implementations.

4.2 Main Results

Table 1 shows that SPRiF attains the highest accuracy in the comparison on S-MNIST, PS-MNIST, and QTDB. The

Table 1: Test accuracy (%) across five temporal benchmarks. Bold and underlined entries denote the best and second-best reported values in each column. A dash indicates that the corresponding paper did not report that benchmark in our comparison.

Method	S-MNIST	PS-MNIST	QTDB	GSC	SHD
GLIF (Fang et al. 2022)	96.64	90.47	—	—	—
ASRNN (Yao et al. 2021)	98.70	94.30	85.90	92.10	90.40
DH-SNN (Zheng et al. 2024)	98.90	94.52	86.35	93.80	91.34
BRF (Higuchi et al. 2024)	99.10	95.20	<u>87.00</u>	—	91.70
TC-LIF (Yin et al. 2024)	<u>99.20</u>	<u>95.36</u>	—	94.84	88.91
SE-adLIF (Baronig et al. 2025)	—	—	86.88	—	—
SNN-SFA (Salaj et al. 2021)	—	—	—	91.21	—
RadLIF (Kamata et al. 2022)	—	—	—	94.51	—
Heterogeneous SNN (Lange et al. 2021)	—	—	—	—	82.50
MPS-SNN (Ding et al. 2025b)	—	—	—	—	91.19
DGN (Bai, Wang, and Yu 2026)	—	—	—	—	87.78
SPRiF (ours)	99.28	95.86	88.43	<u>94.55</u>	<u>91.52</u>

Table 2: Mechanism ablations (accuracy, %). Each component contributes on all three datasets; merging memory and discharge causes the largest degradation.

Dataset	Full	$\omega=0$	Merged	$\lambda=0$
PS-MNIST	95.82	92.29	93.44	95.32
GSC	94.55	93.83	90.05	94.22
QTDB	88.43	87.78	83.39	87.40

gain is small on S-MNIST (0.08 points) but increases to 0.50 points over the strongest TC-LIF configuration on PS-MNIST and 1.43 points over BRF on QTDB. On GSC and SHD, SPRiF trails the strongest reported result by 0.29 and 0.18 points, respectively. This pattern does not support a universal superiority claim; it shows that the same neuron mechanism remains competitive across pixel sequences, physiological signals, frame-based speech, and event streams.

The comparison is also parameter-efficient. SPRiF uses 0.067M parameters on S-MNIST and PS-MNIST, 1.77k on QTDB, 0.13M on GSC, and 0.05M on SHD. Relative to the strongest accuracy baseline in each column, this is approximately 55% fewer parameters on both MNIST variants, comparable on QTDB (1.77k vs. 1.73k), 35% fewer on GSC, and 55% fewer on SHD. Thus, the multi-state neuron does not require a larger end-to-end network to achieve the reported accuracy.

4.3 Mechanism Ablations

We isolate the three design choices on PS-MNIST, GSC, and QTDB. Ablation A sets $\omega = 0$, removing rotational coupling while retaining the real decay path. Ablation B merges memory and discharge into a single three-dimensional state whose first coordinate is the membrane and reset target. Ablation C fixes $\lambda = 0$, reducing Eq. (8) to scalar reset. All variants otherwise retain the task-specific training configuration.

Table 2 gives two consistent findings. First, merging the states reduces accuracy by 2.38–5.04 points, directly supporting functional decomposition rather than state dimensionality alone. Second, rotation contributes 0.65–3.53 points,

with the largest effect on the long permuted sequence. Projective reset gives a smaller but consistent gain of 0.33–1.03 points. The ordering is informative: separating memory from discharge is the primary mechanism, while spectral rotation and a learnable reset direction refine how the two subsystems process temporal structure.

4.4 Dynamical Analysis

The equations guarantee that reset does not directly modify $\mathbf{x}_t$; we additionally inspect trained trajectories to determine whether the resulting dynamics are visible around spikes. Figure 1 aligns a representative pSMNIST neuron around a spike. Its slow coordinates evolve continuously through the event, whereas the membrane coordinate shows the characteristic post-spike correction. This observation verifies that the implementation follows the intended state partition rather than merely naming two groups of variables.

We next inject a unit impulse directly into trained neurons and record 100 steps of slow-state evolution. Figure 2 samples neurons from PS-MNIST, GSC, and QTDB. Learned α values span rapid to persistent real traces, while the rotational coordinates exhibit different frequencies and damping rates. The gallery therefore provides a mechanistic counterpart to the rotation ablation: the constrained spectral state is used to form heterogeneous temporal kernels rather than collapsing to one shared exponential response.

Finally, the learned reset coefficients are distributed on both sides of zero in all three analyzed models rather than collapsing uniformly to scalar reset. This is consistent with the positive, albeit smaller, effect of the $\lambda = 0$ ablation. We avoid interpreting individual signs biologically: the evidence supports learned directional diversity in the fast state, not a universal semantic meaning for positive or negative λ .

5 Limitations

SPRiF trades one scalar state for five state values per neuron, increasing activation memory and neuron-local arithmetic even when the end-to-end parameter count is small. The reported benchmark numbers are single runs and several baselines are taken from their original papers under task-specific

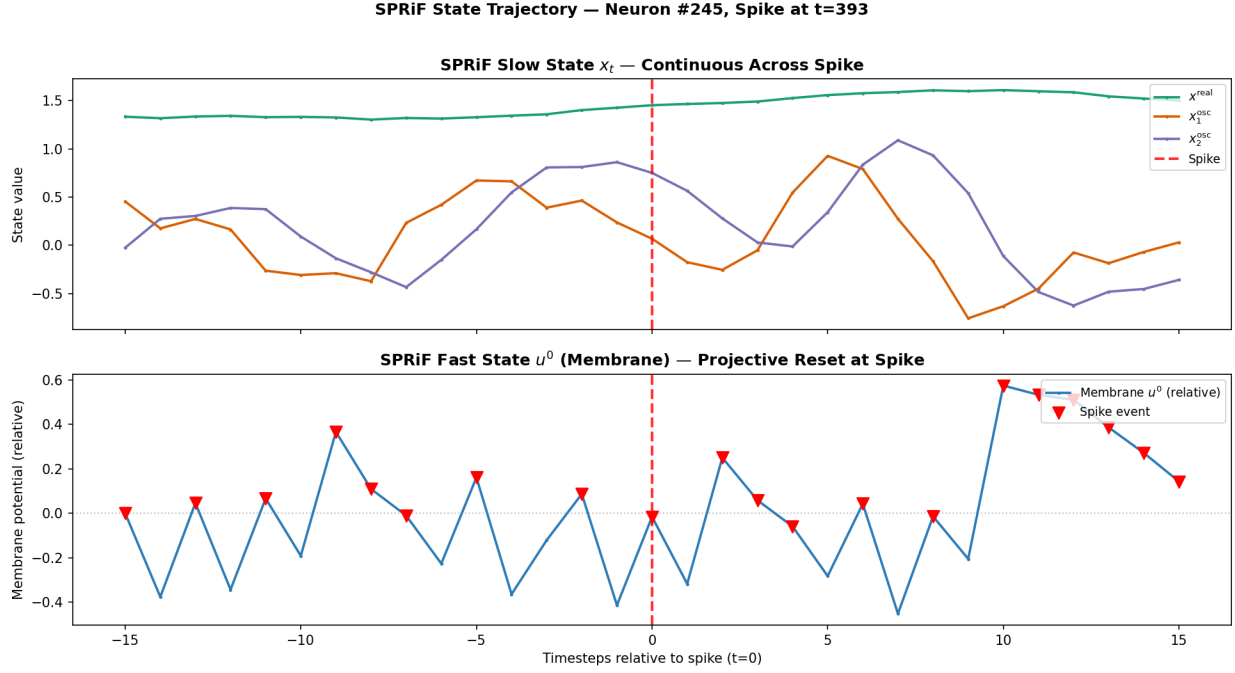


Figure 1: State trajectories of a trained SPRiF neuron around a spike. The three slow coordinates (top) remain continuous at the aligned event, while the fast membrane coordinate (bottom) exhibits spike-triggered corrections.

protocols; repeated-seed statistics and a unified reimplementation would be required for significance claims. The current evaluation focuses on small- to medium-scale temporal classification and does not establish energy gains on neuromorphic hardware. Finally, the diagnostic analyses demonstrate state continuity and diverse kernels, but do not prove that every learned mode is necessary for every input. These boundaries motivate repeated-seed evaluation, matched wall-clock and spike-operation measurements, and larger long-context benchmarks.

6 Conclusion

SPRiF reorganizes a spiking neuron around a simple principle: the state that stores temporal context need not be the state that is reset after a spike. A constrained spectral slow state supplies real and rotational memory modes, while a fast discharge state produces binary spikes and absorbs a learnable projective reset. Across five temporal benchmarks, this design is strongest in the reported comparison on three tasks and competitive on the other two with modest network size. Ablations identify slow/fast separation as the largest contributor, with consistent additional gains from rotation and projective reset; trajectory and impulse analyses make these mechanisms directly inspectable. The results position functional state decomposition as a reusable neuron-level design direction, while broader-scale and hardware evaluation remain open.

SPRiF Impulse Response Gallery — Learned Temporal Kernels

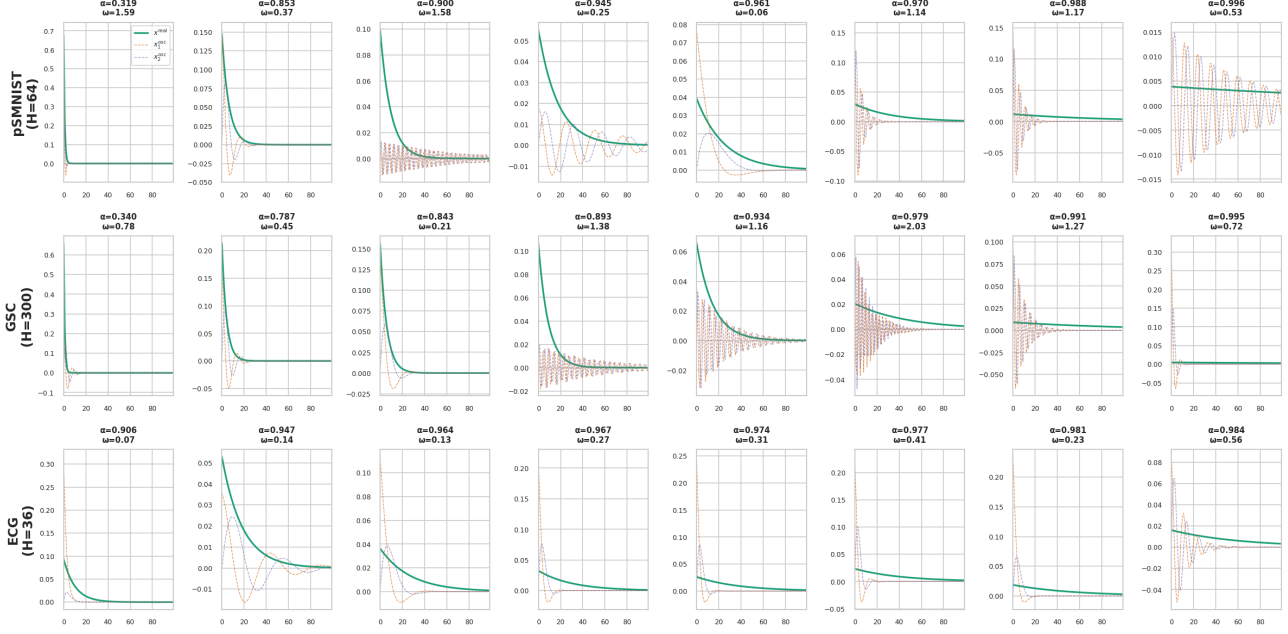


Figure 2: Impulse responses of sampled neurons from PS-MNIST, GSC, and QADB models. Solid curves show the real decay coordinate; dashed curves show the two rotational coordinates. Learned parameters produce heterogeneous decay horizons and oscillatory responses.

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